



NCI Protocol #: 10323

Patient ID:

Date:

1 of 19

Patient Name:

Date Collected:

Biopsy Site: **Liver**

Sample ID:

Telephone:

Tumor Content (%): **20_49%**

Patient DOB:

Referring Physician:

Primary Diagnosis: **Metastatic colorectal adenocarcinoma**

MoCha ID:

Fax:

Preanalytics to include tumor enrichment as required is performed by Van Andel Research Institute; Grand Rapids; MI 49503

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Relevant Colorectal Cancer Findings

Gene	Finding
BRAF	None detected
KRAS	None detected
NRAS	NRAS wild type
NTRK1	None detected
NTRK2	None detected
NTRK3	None detected

Relevant Biomarkers

Tier	Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
	NRAS wild type	panitumumab¹ panitumumab + chemotherapy¹	-	-
	Prognostic significance: None Diagnostic significance: None			

Public data sources included in relevant therapies: FDA¹, NCCN

Public data sources included in prognostic and diagnostic significance: NCCN

Tier Reference: Li et al. *Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists.* J Mol Diagn. 2017 Jan;19(1):4-23.

* Includes biosimilars

Relevant Biomarkers (continued)

Tier	Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
IA	ERBB2 amplification	lapatinib + trastuzumab pertuzumab + trastuzumab trastuzumab deruxtecan	ado-trastuzumab emtansine ¹ irbinitinib + trastuzumab + chemotherapy ¹ lapatinib + chemotherapy ¹ lapatinib + hormone therapy ¹ margetuximab + chemotherapy ¹ neratinib ¹ neratinib + chemotherapy ¹ pembrolizumab + trastuzumab + chemotherapy ¹ pertuzumab + trastuzumab + chemotherapy ¹ pertuzumab/trastuzumab/ hyaluronidase-zzxf + chemotherapy ¹ trastuzumab and hyaluronidase-oysk ¹ trastuzumab and hyaluronidase-oysk + chemotherapy ¹ trastuzumab deruxtecan ¹ trastuzumab* ¹ trastuzumab* + chemotherapy ¹ hormone therapy lapatinib + trastuzumab lapatinib + trastuzumab + hormone therapy pertuzumab + trastuzumab pertuzumab + trastuzumab + hormone therapy + chemotherapy trastuzumab + hormone therapy trastuzumab + hormone therapy + chemotherapy	57
	Prognostic significance: None Diagnostic significance: None			
III	PIK3CB F155C	None	None	6
	Prognostic significance: None Diagnostic significance: None			

Public data sources included in relevant therapies: FDA¹, NCCN
Public data sources included in prognostic and diagnostic significance: NCCN
Tier Reference: Li et al. Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists. J Mol Diagn. 2017 Jan;19(1):4-23.
* Includes biosimilars

Variant Details

DNA Sequence Variants							
Gene	Amino Acid Change	Coding	Variant ID	Locus	Allele Frequency	Transcript	Variant Effect
PIK3CB	p.(F155C)	c.464T>G	.	chr3:138461557	27.92%	NM_006219.2	missense

Variant Details (continued)

Copy Number Variations

Gene	Locus	Copy Number
ERBB2	chr17:37868167	37.03

Comments:

N/A

Signed By: _____

Biomarker Descriptions

ERBB2 (erb-b2 receptor tyrosine kinase 2)

Background: The ERBB2 gene encodes the erb-b2 receptor tyrosine kinase 2, a member of the human epidermal growth factor receptor (HER) family. Along with ERBB2/HER2, EGFR/ERBB1/HER1, ERBB3/HER3, and ERBB4/HER4 make up the HER protein family¹. All ERBB/HER proteins encode transmembrane receptor tyrosine kinases. However, ERBB2/HER2 is an orphan receptor with no known ligand. ERBB2 preferentially binds other ligand bound ERBB/HER family members to form hetero-dimers resulting in the activation of ERBB2 tyrosine kinase activity and subsequent activation of the PI3K/AKT/MTOR and RAS/RAF/MAPK/ERK signaling pathways which promote cell proliferation, differentiation, and survival². Recurrent focal amplification of the ERBB2 gene leads to increased expression in several cancer types. ERBB2 overexpression in immortalized cell lines is oncogenic and leads to ERBB2 homo-dimerization and activation without ligand binding^{3,4,5}.

Alterations and prevalence: ERBB2 gene amplification occurs in 10-20% of breast, esophageal, and gastric cancers, 5-10% of bladder, cervical, pancreas, and uterine cancers, and 1-5% of colorectal, lung, and ovarian cancers^{6,7,8,9,10,11,12,13}. Recurrent somatic activating mutations in ERBB2/HER2 occur at low frequencies (<1%) in diverse cancer types^{13,14,15}. In breast, bladder, and colorectal cancers, the most common recurrent ERBB2 activating mutations include kinase domain mutations L755S and V777L and the extracellular domain mutation S310F. In lung cancer, the most common recurrent ERBB2 activating mutations include in-frame exon 20 insertions, particularly Y772_A775dup.

Potential relevance: The discovery of ERBB2/HER2 as an important driver of breast cancer in 1987 led to the development of trastuzumab, a humanized monoclonal antibody with specificity to the extracellular domain of HER2^{16,17}. Trastuzumab¹⁸ was FDA approved for the treatment of HER2 positive breast cancer in 1998, and subsequently in HER2 positive metastatic gastric and gastroesophageal junction adenocarcinoma in 2010. Additional monoclonal antibody therapies have been approved by the FDA for HER2-positive breast cancer including pertuzumab¹⁹ (2012), a humanized monoclonal antibody that inhibits HER2 dimerization, and ado-trastuzumab emtansine²⁰ (2013), a conjugate of trastuzumab and a potent antimicrotubule agent. The combination of pertuzumab, trastuzumab, and a taxane is the preferred front-line regimen for HER2-positive metastatic breast cancer²¹. In addition to monoclonal antibodies, the small molecule inhibitor lapatinib²², with specificity for both EGFR and ERBB2, was FDA approved (2007) for the treatment of patients with advanced HER2-positive breast cancer who have received prior therapy including trastuzumab. In 2017, the FDA approved the use of neratinib²³, an irreversible kinase inhibitor of EGFR, ERBB2/HER2, and ERBB4, for the extended

Biomarker Descriptions (continued)

adjuvant treatment of adult patients with early stage HER2-positive breast cancer. In 2020, the FDA approved neratinib²³ in combination with capecitabine for HER2-positive advanced or metastatic patients after two or more prior HER2-directed therapies. Also in 2020, the TKI irbinetinib²⁴ was FDA approved for HER2 overexpressing or amplified breast cancer in combination with trastuzumab and capecitabine. In 2021, the PD-1 blocking antibody, pembrolizumab, in combination with trastuzumab, fluoropyrimidine- and platinum-based chemotherapy, was approved for HER2 amplified gastric or gastroesophageal (GEJ) adenocarcinoma in the first line²⁵. The vaccine, nelipepimut-S²⁶, was granted fast-track designation by the FDA (2016) in patients with low to intermediate HER2 expressing (IHC score 1+ or 2+) breast cancer. In 2018 fast-track designation was granted to the monoclonal antibody margetuximab²⁷ in patients with ERBB2 positive breast cancer previously treated with an anti-HER2 therapy. In 2019, fast track designation was granted to the HER2-targeting antibody drug conjugate, amcenestrant²⁸, for HER2-positive advanced or metastatic breast cancer after one or more prior anti-HER2 based regimens. Additionally, in 2019, the novel bispecific antibody, zanidatamab²⁹, received fast-track designation in combination with standard chemotherapy for patients with HER2-overexpressing gastroesophageal adenocarcinoma (GEA) and breakthrough therapy designation (2020) as a monotherapy for patients with HER2-amplified biliary tract cancer³⁰. In 2020, BDTX-189³¹ received fast-track designation for adult patients with solid tumors harboring an allosteric human ERBB2 mutation or exon 20 insertion, and the humanized anti-HER2 antibody drug conjugate disitamab vedotin received breakthrough designation for adult patients with HER2-positive urothelial cancer after previous platinum-chemotherapy treatment³². In 2021, the antibody-drug conjugate ARX788³³ received fast-track designation as a monotherapy for advanced or metastatic HER2-positive breast cancer that have progressed on one or more anti-HER2 regimens. Additionally, in 2021, fast track designation was granted to HER2 targeted chimeric antigen receptor macrophage (CAR-M), CT-0508³⁴, for HER2-overexpressing solid tumors. Certain activating mutations have been observed to impart sensitivity to neratinib, afatinib, lapatinib, and trastuzumab, or dacomitinib in early and ongoing clinical studies^{35,36,37,38,39}. ERBB2 kinase domain mutations R896G and V659E both showed response to afatinib in two NSCLC case studies^{40,41}. Additionally, acquired HER2 mutations in estrogen receptor-positive (ER+) breast cancer have been shown to confer resistance to hormone therapy⁴². However, this was shown to be overcome by neratinib in combination with therapies targeting ER⁴².

NRAS (NRAS proto-oncogene, GTPase)

Background: The NRAS proto-oncogene encodes a GTPase that functions in signal transduction and is a member of the RAS superfamily which also includes KRAS and HRAS. RAS proteins mediate the transmission of growth signals from the cell surface to the nucleus via the PI3K/AKT/MTOR and RAS/RAF/MEK/ERK pathways, which regulate cell division, differentiation, and survival^{43,44,45}.

Alterations and prevalence: Recurrent mutations in RAS oncogenes cause constitutive activation and are found in 20-30% of cancers. NRAS mutations are particularly common in melanomas (up to 25%) and are observed at frequencies of 5-10% in acute myeloid leukemia, colorectal, and thyroid cancers^{12,46}. The majority of NRAS mutations consist of point mutations at G12, G13, and Q61^{12,47}. Mutations at A59, K117, and A146 have also been observed but are less frequent^{13,48}.

Potential relevance: Currently, no therapies are approved for NRAS aberrations. The EGFR antagonists, cetuximab⁴⁹ and panitumumab⁵⁰, are contraindicated for treatment of colorectal cancer patients with NRAS mutations in exon 2 (codons 12 and 13), exon 3 (codons 59 and 61), and exon 4 (codons 117 and 146)⁴⁸. NRAS mutations are associated with poor prognosis in patients with low-risk myelodysplastic syndrome⁵¹ as well as melanoma⁵². In a phase III clinical trial in patients with advanced NRAS-mutant melanoma, binimetinib improved progression free survival (PFS) relative to dacarbazine with median PFS of 2.8 and 1.5 months, respectively⁵³.

PIK3CB (phosphatidylinositol-4,5-bisphosphate 3-kinase catalytic subunit beta)

Background: The PIK3CB gene encodes the phosphatidylinositol-4,5-bisphosphate 3-kinase catalytic subunit beta of the class I phosphatidylinositol 3-kinase (PI3K) enzyme⁵⁴. PI3K is a heterodimer that contains a p85 regulatory subunit, which couples one of four p110 catalytic subunits to activated tyrosine protein kinases^{54,55}. The p110 catalytic subunits include p110α, β, δ, γ and are encoded by genes PIK3CA, PIK3CB, PIK3CD, and PIK3CG, respectively⁵⁴. PI3K catalyzes the conversion of phosphatidylinositol (4,5)-bisphosphate (PI(4,5)P₂) into phosphatidylinositol (3,4,5)-trisphosphate (PI(3,4,5)P₃) while the phosphatase and tensin homolog (PTEN) catalyzes the reverse reaction^{56,57}. The reversible phosphorylation of inositol lipids regulate diverse aspects of cell growth and metabolism^{56,57,58,59}. Aberrations in PIK3CB that lead to activation of the PI3K/AKT/MTOR pathway have been observed to promote tumor formation, suggesting an oncogenic role for PIK3CB^{54,60,61}.

Alterations and prevalence: Somatic mutations in PIK3CB are predominantly missense with amino acid substitutions at D1067 being the most recurrent and observed to lead to hyperactivation of the PI3K pathway^{12,13,62}. PIK3CB mutations are observed in about 9% of uterine cancer and 2-3% of melanoma, glioblastoma, cholangiocarcinoma, colorectal, bladder, stomach, esophageal, and squamous

Biomarker Descriptions (continued)

lung cancers^{12,13}. Amplification of PIK3CB is also observed in 9% of squamous lung cancer, 7% of cervical cancer, and 5-6% of head and neck, ovarian, and esophageal cancers^{12,13}.

Potential relevance: Currently, no therapies are approved for PIK3CB aberrations.

Relevant Therapy Summary

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

NRAS wild type

Relevant Therapy	FDA	NCCN	Clinical Trials*
panitumumab	☒	☒	☒
panitumumab + FOLFOX	☒	☒	☒

ERBB2 amplification

Relevant Therapy	FDA	NCCN	Clinical Trials*
trastuzumab deruxtecan	☐	☒	☒ (II)
ado-trastuzumab emtansine	☐	☐	☒ (II)
trastuzumab	☐	☐	☒ (II)
irbinitinib + trastuzumab + capecitabine	☐	☐	☒
lapatinib + capecitabine	☐	☐	☒
neratinib	☐	☐	☒
neratinib + capecitabine	☐	☐	☒
pertuzumab + trastuzumab + chemotherapy	☐	☐	☒
pertuzumab + trastuzumab + docetaxel	☐	☐	☒
trastuzumab + capecitabine + cisplatin	☐	☐	☒
trastuzumab + carboplatin + docetaxel	☐	☐	☒
trastuzumab + cisplatin + fluorouracil	☐	☐	☒
trastuzumab + docetaxel	☐	☐	☒
trastuzumab + paclitaxel	☐	☐	☒
lapatinib + letrozole	☐	☒	☒

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

ERBB2 amplification (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
margetuximab + chemotherapy	☐	☒	☒
pembrolizumab + trastuzumab + chemotherapy + fluoropyrimidine	☐	☒	☒
pertuzumab/trastuzumab/hyaluronidase-zzxf + cyclophosphamide + doxorubicin	☐	☒	☒
pertuzumab/trastuzumab/hyaluronidase-zzxf + docetaxel	☐	☒	☒
trastuzumab (Biocon)	☐	☒	☒
trastuzumab (Biocon) + capecitabine + cisplatin	☐	☒	☒
trastuzumab (Biocon) + carboplatin + docetaxel	☐	☒	☒
trastuzumab (Biocon) + cisplatin + fluorouracil	☐	☒	☒
trastuzumab (Biocon) + docetaxel	☐	☒	☒
trastuzumab (Biocon) + paclitaxel	☐	☒	☒
trastuzumab (Celltrion)	☐	☒	☒
trastuzumab (Celltrion) + capecitabine + cisplatin	☐	☒	☒
trastuzumab (Celltrion) + carboplatin + docetaxel	☐	☒	☒
trastuzumab (Celltrion) + cisplatin + fluorouracil	☐	☒	☒
trastuzumab (Celltrion) + docetaxel	☐	☒	☒
trastuzumab (Celltrion) + paclitaxel	☐	☒	☒
trastuzumab (Pfizer)	☐	☒	☒
trastuzumab (Pfizer) + capecitabine + cisplatin	☐	☒	☒
trastuzumab (Pfizer) + carboplatin + docetaxel	☐	☒	☒
trastuzumab (Pfizer) + cisplatin + fluorouracil	☐	☒	☒
trastuzumab (Pfizer) + docetaxel	☐	☒	☒
trastuzumab (Pfizer) + paclitaxel	☐	☒	☒
trastuzumab (Samsung Bioepis)	☐	☒	☒
trastuzumab (Samsung Bioepis) + capecitabine + cisplatin	☐	☒	☒
trastuzumab (Samsung Bioepis) + carboplatin + docetaxel	☐	☒	☒
trastuzumab (Samsung Bioepis) + cisplatin + fluorouracil	☐	☒	☒

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

ERBB2 amplification (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
trastuzumab (Samsung Bioepis) + docetaxel	☐	☒	☒
trastuzumab (Samsung Bioepis) + paclitaxel	☐	☒	☒
trastuzumab and hyaluronidase-oysk	☐	☒	☒
trastuzumab and hyaluronidase-oysk + carboplatin + docetaxel	☐	☒	☒
trastuzumab and hyaluronidase-oysk + cyclophosphamide + doxorubicin + paclitaxel	☐	☒	☒
trastuzumab and hyaluronidase-oysk + docetaxel	☐	☒	☒
trastuzumab and hyaluronidase-oysk + paclitaxel	☐	☒	☒
pertuzumab + trastuzumab	☒	☒	☒ (II)
lapatinib + trastuzumab	☒	☒	☒
aromatase inhibitor	☒	☐	☒
fulvestrant	☒	☐	☒
hormone therapy	☒	☐	☒
lapatinib + aromatase inhibitor	☒	☐	☒
lapatinib + trastuzumab + aromatase inhibitor	☒	☐	☒
margetuximab + capecitabine	☒	☐	☒
margetuximab + eribulin	☒	☐	☒
margetuximab + gemcitabine	☒	☐	☒
margetuximab + vinorelbine	☒	☐	☒
neratinib + paclitaxel	☒	☐	☒
pembrolizumab + trastuzumab + capecitabine + cisplatin	☒	☐	☒
pembrolizumab + trastuzumab + capecitabine + oxaliplatin	☒	☐	☒
pembrolizumab + trastuzumab + cisplatin + fluorouracil	☒	☐	☒
pembrolizumab + trastuzumab + fluorouracil + oxaliplatin	☒	☐	☒
pertuzumab + trastuzumab + carboplatin + docetaxel	☒	☐	☒
pertuzumab + trastuzumab + hormone therapy + chemotherapy	☒	☐	☒
pertuzumab + trastuzumab + paclitaxel	☒	☐	☒

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

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Disclaimer: The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2022.03(006).

Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

ERBB2 amplification (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
tamoxifen	×	○	×
trastuzumab + aromatase inhibitor	×	○	×
trastuzumab + capecitabine	×	○	×
trastuzumab + capecitabine + oxaliplatin	×	○	×
trastuzumab + carboplatin + docetaxel + fluorouracil	×	○	×
trastuzumab + carboplatin + paclitaxel	×	○	×
trastuzumab + chemotherapy	×	○	×
trastuzumab + chemotherapy (other)	×	○	×
trastuzumab + cisplatin + docetaxel	×	○	×
trastuzumab + cisplatin + docetaxel + fluorouracil	×	○	×
trastuzumab + cisplatin + paclitaxel	×	○	×
trastuzumab + cyclophosphamide + docetaxel	×	○	×
trastuzumab + docetaxel + fluorouracil + oxaliplatin	×	○	×
trastuzumab + fluorouracil	×	○	×
trastuzumab + fluorouracil + irinotecan	×	○	×
trastuzumab + fluorouracil + oxaliplatin	×	○	×
trastuzumab + fulvestrant	×	○	×
trastuzumab + hormone therapy + chemotherapy	×	○	×
trastuzumab + tamoxifen	×	○	×
trastuzumab + vinorelbine	×	○	×
pyrotinib	×	×	● (IV)
ado-trastuzumab emtansine + atezolizumab, pertuzumab/trastuzumab/hyaluronidase-zzxf, pertuzumab/trastuzumab/hyaluronidase-zzxf + chemotherapy, ado-trastuzumab emtansine + irbinetinib	×	×	● (II)
atezolizumab + pertuzumab + trastuzumab	×	×	● (II)
KN026, KN046	×	×	● (II)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

ERBB2 amplification (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
pertuzumab + trastuzumab, atezolizumab + pertuzumab/trastuzumab/hyaluronidase-zzxf, irbinitinib + trastuzumab	×	×	● (II)
pertuzumab, cetuximab, trastuzumab, chemotherapy	×	×	● (II)
pyrotinib, trastuzumab	×	×	● (II)
trastuzumab, cetuximab, neratinib	×	×	● (II)
trastuzumab, chemotherapy	×	×	● (II)
trastuzumab, pertuzumab, ado-trastuzumab emtansine, lapatinib	×	×	● (II)
zanidatamab, chemotherapy, bevacizumab	×	×	● (II)
ARX-788	×	×	● (I/II)
BDC-1001, nivolumab	×	×	● (I/II)
BDTX-189	×	×	● (I/II)
CART-HER2 (Bellicum Pharmaceuticals), rimiducid	×	×	● (I/II)
E01001	×	×	● (I/II)
irbinitinib, trastuzumab, pembrolizumab, chemotherapy	×	×	● (I/II)
MRG-002	×	×	● (I/II)
SBT6050, trastuzumab deruxtecan, irbinitinib, trastuzumab, chemotherapy	×	×	● (I/II)
TAC01-HER2	×	×	● (I/II)
zanidatamab, evorpaccept	×	×	● (I/II)
zotatifin	×	×	● (I/II)
ado-trastuzumab emtansine (Shanghai Fosun Pharma)	×	×	● (I)
AMX-3009	×	×	● (I)
anti-HER-2 MAb (Anke Biotechnology)	×	×	● (I)
BAY-2701439	×	×	● (I)
CART	×	×	● (I)
CART-HER2	×	×	● (I)
CART-HER2, chemotherapy	×	×	● (I)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

ERBB2 amplification (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
ceralasertib, trastuzumab deruxtecan	×	×	● (I)
DP-303c	×	×	● (I)
GQ1001	×	×	● (I)
M802	×	×	● (I)
MBS301	×	×	● (I)
MT-5111	×	×	● (I)
neratinib, palbociclib, everolimus, trametinib	×	×	● (I)
pyrotinib, SHR-6390	×	×	● (I)
S-095029, Sym-021, antiHER2 therapy	×	×	● (I)
SAR-443216	×	×	● (I)
SBT6050, pembrolizumab, cemiplimab	×	×	● (I)
SHR-A1811	×	×	● (I)
SPH5030	×	×	● (I)
TAS 2940	×	×	● (I)
trastuzumab deruxtecan, olaparib	×	×	● (I)
trastuzumab duocarmazine, chemotherapy	×	×	● (I)
zanidatamab	×	×	● (I)
ZN-A-1041	×	×	● (I)

PIK3CB F155C

Relevant Therapy	FDA	NCCN	Clinical Trials*
atezolizumab + ipatasertib, atezolizumab + ipatasertib + chemotherapy	×	×	● (II)
ipatasertib	×	×	● (II)
samotolisib	×	×	● (II)
ipatasertib, atezolizumab	×	×	● (I/II)
capiasertib, midazolam	×	×	● (I)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

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Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

PIK3CB F155C (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
paxalisib, radiation therapy	×	×	☒ (I)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Clinical Trials Summary

ERBB2 amplification

NCT ID	Title	Phase
No NCT ID	An Exploratory Basket Study of Non Breast Cancer Patients with HER2 Mutation or Amplification who Failed in Standard Treatment with Pyrotinib Maleate Tablets	IV
NCT04744831	A Phase II, Multicenter, Randomized, Study of Trastuzumab Deruxtecan in Participants With HER2-overexpressing Locally Advanced, Unresectable or Metastatic Colorectal Cancer (DESTINY-CRC02)	II
NCT03457896	A Phase II Study Evaluating the Combination of Neratinib Plus Trastuzumab or Neratinib Plus Cetuximab in Patients With "Quadruple Wild-Type" (KRAS/NRAS/BRAF/PIK3CA Wild-Type) Metastatic Colorectal Cancer Based on HER2 Status: Amplified, Non-Amplified (Wild-Type) or Mutated	II
NCT03185988	Multicenter, Phase II Study of Chemotherapy in Combination With Trastuzumab in Patients of Pretreated, HER2 Positive, Relapse or Metastatic Carcinoma of Digestive System	II
NCT04591431	The Rome Trial From Histology to Target: the Road to Personalize Target Therapy and Immunotherapy	II
NCT03929666	Phase II Study of ZW25 Plus First-line Combination Chemotherapy in HER2-Expressing Gastrointestinal (GI) Cancers, Including Gastroesophageal Adenocarcinoma (GEA), Biliary Tract Cancer (BTC), and Colorectal Cancer (CRC)	II
NCT04650451	A Phase I/II, Open-Label, Multicenter, Non-Randomized, Safety and Activity Study of HER2-Targeted Dual Switch CAR-T Cells (BPX-603) In Subjects With Previously Treated Advanced HER2-Positive Solid Tumors	I/II
NCT05091528	An Open-label, Phase I/II, Dose-escalation and Expansion Study of SBT6050 Combined With Other HER2-directed Therapies in Subjects With Pretreated Unresectable Locally Advanced and/or Metastatic HER2-expressing or HER2-amplified Cancers	I/II
NCT04040699	A Phase Ib Study to Evaluate Efficacy, Safety and Tolerability of KN026 Combined With KN046 in Subjects With HER2 Positive Solid Tumor.	I
NCT04460456	A Phase I/IB, Open-Label, Dose Escalation and Expansion Study of SBT6050 Alone and in Combination With PD-1 Inhibitors in Subjects With Advanced Solid Tumors Expressing HER2	I
NCT02892123	Phase I Trial of ZW25 in Patients With Locally Advanced (Unresectable) and/or Metastatic HER2-expressing Cancers.	I
No NCT ID	TR Study of "Multicenter Phase II Study to Evaluate Efficacy and Safety of Combination Therapy with Trastuzumab and Pertuzumab in Patients with HER2-Positive Metastatic Colorectal Cancer"	II

Clinical Trials Summary (continued)

ERBB2 amplification (continued)

NCT ID	Title	Phase
NCT03843749	Pyrotinib in Combination With Trastuzumab in Treatment-Refractory, HER2-Positive Metastatic Colorectal Cancer.	II
NCT04430738	A Phase Ib/II Dose Escalation and Expansion Study of Tucatinib in Combination With Trastuzumab and Oxaliplatin-based Chemotherapy or Pembrolizumab-containing Combinations for HER2+ Gastrointestinal Cancers	I/II
NCT04602117	ISPY-P1.01: Evaluating the Safety of Weekly Paclitaxel With Trastuzumab Duocarmazine (SYD985) in Patients With Metastatic Cancer: A Phase I/Ib Trial	I
No NCT ID	Single Arm, Open Label, Signal Seeking, Phase IIa Trial Of The Activity Of Trastuzumab Emtansine (T-DM1) In Patients With Tumours Harboursing HER2 Amplifications Or Mutations	II
NCT04632992	MyTACTIC: An Open-Label Phase II Study Evaluating Targeted Therapies in Patients Who Have Advanced Solid Tumors With Genomic Alterations or Protein Expression Patterns Predictive of Response	II
NCT04551521	Continuous ReAssessment With Flexible ExTension in Rare Malignancies - CRAFT: The NCT-PMO-1602 Phase II Trial	II
NCT02693535	Targeted Agent and Profiling Utilization Registry (TAPUR) Study.	II
NCT03365882	S1613, A Randomized Phase II Study of Trastuzumab and Pertuzumab (TP) Compared to Cetuximab and Irinotecan (CETIRI) in Advanced/Metastatic Colorectal Cancer (mCRC) With HER-2 Amplification	II
NCT04423185	Platform Study of Genotyping Guided Precision Medicine for Rare Tumors in China	II
NCT03239015	Efficacy and Safety of Targeted Precision Therapy in Refractory Tumor With Druggable Molecular Event	II
NCT04482309	A Phase II, Multicenter, Open-label Study to Evaluate the Efficacy and Safety of Trastuzumab Deruxtecan (T-DXd, DS-8201a) for the Treatment of Selected HER2 Expressing Tumors (DESTINY-PanTumor02)	II
No NCT ID	A Single-Arm, Open, Basket, Phase Ib/II Clinical Study To Evaluate The Efficacy And Safety Of Recombinant Humanized Anti-HER2 Monoclonal Antibody-AS269 Conjugate (ARX788) In Patients With HER2-Positive Advanced Solid Tumors	I/II
NCT04278144	Phase I/II Study of BDC-1001 as a Single Agent and in Combination With Nivolumab in Patients With Advanced HER2-Expressing Solid Tumors	I/II
NCT04209465	MasterKey-01: A Phase I/II, Open-label, Two-part, Multicenter Study to Assess the Safety, Tolerability, Pharmacokinetics & Antitumor Activity of BDTX-189, an Inhibitor of Allosteric ErbB Mutations, in Patients w/ Advanced Solid Malignancies.	I/II
NCT04727151	A Phase I/II Study Investigating Safety and Efficacy of Autologous TAC T Cells Targeting HER2 in Relapsed or Refractory Solid Tumors	I/II
NCT05027139	A Phase Ib/II, 2-part Open-label Study to Assess the Safety and Antitumor Activity of Zanidatamab in Combination With ALX148 in Advanced HER2-expressing Cancer	I/II
NCT04092673	A Phase 1-2 Dose-Escalation and Cohort-Expansion Study of Intravenous Zotatfin (eFT226) in Subjects With Selected Advanced Solid Tumor Malignancies	I/II

Clinical Trials Summary (continued)

ERBB2 amplification (continued)

NCT ID	Title	Phase
NCT03944499	A Phase I, Multicenter, Open-label, Single-arm Study: A Dose-escalation Phase (Phase 1a) Evaluating FS-1502 in Patients With HER2 Expressed Advanced Solid Tumors; and a Dose-expanded Cohort (Phase 1b) Evaluating FS-1502 in Patients With Local Advanced or Metastatic, HER2 Positive Breast Cancer.	I
No NCT ID	Single-Arm, Open-Label, Single-Dose And Multiple-Dose Phase Ia Clinical Study Of Tolerability And Pharmacokinetics Of AMX3009 In Patients With HER2-Positive Advanced Solid Tumors	I
No NCT ID	A Phase I Clinical Study To Evaluate The Safety, Tolerability And Pharmacokinetic Characteristics of Recombinant Anti-HER2 Humanized HuA21 Monoclonal Antibody Injection in the Treatment of HER2-Positive Advanced Malignant Solid Tumors	I
NCT03255070	A Phase I, Multicenter, Open-label, Multiple Dose-escalation and Expansion Study of ARX788, as Monotherapy in Advanced Solid Tumors With HER2 Expression	I
NCT04147819	A Phase I Open-label, First-in-human, Multi-center Study to Evaluate the Safety, Tolerability, Pharmacokinetics and Anti-tumor Activity of Thorium-227 Labeled Antibody-chelator Conjugate BAY2701439, in Participants With Advanced HER2-expressing Tumors	I
NCT04511871	A Phase I Trial to Assess Safety, Tolerability and Anti-tumor Activity of Autologous T Cell Modified Chimeric Antigen Receptor (CAR) (CCT303-406) in Patients With Relapsed or Refractory HER2 Positive Solid Tumors	I
NCT02442297	Phase I Study of Intracranial Injection of T Cells Expressing HER2-specific Chimeric Antigen Receptors (CAR) in Subjects With HER2-Positive Tumors of the Central Nervous System (iCAR)	I
NCT03696030	A Phase I Cellular Immunotherapy Study of Intraventricularly Administered Autologous HER2-Targeted Chimeric Antigen Receptor (HER2-CAR) T Cells in Patients With Brain and/or Leptomeningeal Metastases From HER2 Positive Cancers	I
NCT04704661	Phase I/I B Study of DS-8201a in Combination With ATR Inhibition (AZD6738) in Advanced Solid Tumors With HER2 Expression (DASH Trial)	I
NCT04146610	A Phase Ia, Multicenter, Open and Dose-increasing Study of DP303c to Evaluate the Safety , Pharmacokinetics, Immunogenicity and Antitumor Activity of Subjects With HER2-Positive Advanced Solid Tumors	I
NCT04450732	A Phase I, First-In-Human, Multicenter, Open-Label, Study of GQ1001, a HER2 Targeted Antibody-Drug Conjugate, Administered Intravenously, in Adult Patients With HER2-Positive Advanced Solid Tumors	I
NCT04501770	A Phase I Study to Evaluate the Safety, Tolerability, Pharmacokinetics, Pharmacodynamics and Immunogenicity Profiles of the Recombinant Anti-HER2 and Anti-CD3 Humanized Bispecific Antibody (M802) in HER2-Positive Advanced Solid Tumors	I
NCT03842085	Evaluation on the Safety and Pharmacokinetics of Recombinant Humanized Bispecific Monoclonal Antibody MBS301 for Injection in Treatment of HER2 Positive Recurrent or Metastatic Malignant Solid Tumor	I
NCT04029922	A Phase I Open-label, Multicenter Dose Escalation Study of MT-5111 in Subjects With Previously Treated Advanced HER2-positive Solid Tumors	I
NCT03065387	Phase I Study of the Pan-ERBB Inhibitor Neratinib Given in Combination With Everolimus, Palbociclib, or Trametinib in Advanced Cancer Subjects With EGFR Mutation/Amplification, HER2 Mutation/Amplification, or HER3/4 Mutation or KRAS Mutation	I

Clinical Trials Summary (continued)

ERBB2 amplification (continued)

NCT ID	Title	Phase
NCT04585958	A Phase I Study of DS-8201a in Combination With Olaparib in HER2-Expressing Malignancies	I
NCT04487236	A Phase I Clinical Study to Evaluate the Safety, Tolerability, Pharmacokinetics, and Efficacy of ZN-A-1041 Enteric Capsules as a Single Agent or in Combination With Capecitabine Tablets in Patients With HER2-Positive Advanced Solid Tumors	I
NCT04521179	A Phase II, Open-Label, Multi-Center Study to Evaluate Efficacy, Safety and Tolerability of KN026 in Combination With KN046 in Patients With Locally Advanced Unresectable or Metastatic HER2-positive Solid Tumors	II
No NCT ID	An Ascending Single And Multiple Dose Study Of The Safety, Tolerability, Pharmacokinetics And Anti-Tumour Activity Of Once-Daily Oral Treatment With EO1001 In Patients With Advanced Cancer.	I/II
NCT04492488	An Open-Label, Multi-center Phase I/II Dose Escalation and Expansion Study to Assess the Safety, Efficacy and Pharmacokinetics of MRG002 in Patients With HER2-Positive Advanced Solid Tumors and Locally Advanced or Metastatic Gastric/Gastroesophageal Junction (GEJ) Cancer	I/II
NCT04941339	A Phase I, Open-label, Multi-center, First in Human, Dose Escalation and Expansion Study to Assess the Safety, Tolerability, Efficacy and Pharmacokinetics of MRG002 in Patients With HER2 Positive Advanced Solid Tumors.	I
NCT03480256	Phase Id Trial to Evaluate SHR6390 Combined With Pyrotinib in Patients With HER2 Positive Advanced Stage Gastric Carcinoma	I
NCT05162755	A Phase Ia/Ib, Open-label, Multicenter Trial Investigating the Safety, Tolerability, and Preliminary Anti-neoplastic Activity of S095029 (Anti-NKG2A) as Monotherapy and in Combination With Sym021 (Anti-PD-1) in Patients With Advanced Solid Tumor Malignancies Followed by an Expansion Part With Triplet Combinations of S095029 and Sym021 and an Anti-HER2 mAb or Anti-EGFR mAbs (Futuximab/ Modotuximab) in Patients With Metastatic Gastric or Colorectal Cancers	I
NCT04446260	A Phase I Multi-Country, Multi-Center, Open-Label Study to Evaluate the Safety, Tolerability, Pharmacokinetics and Efficacy of SHR-A1811 in HER2 Expressing or Mutated Advanced Malignant Solid Tumor Subjects	I
No NCT ID	A Phase I Clinical Study of SPH5030 tablets for the Treatment of Patients with HER2-positive advanced solid tumor	I
NCT03297606	Canadian Profiling and Targeted Agent Utilization Trial (CAPTUR): A Phase II Basket Trial	II
NCT05013554	A Phase I/Ib Open-label, First-in-human, Single Agent, Dose Escalation and Expansion Study for the Evaluation of Safety, Pharmacokinetics, Pharmacodynamics, and Anti-tumor Activity of SAR443216 in Participants with Relapsed/Refractory HER2 Expressing Solid Tumors.	I
NCT04982926	A Phase I Study of TAS2940 in Patients With Locally Advanced or Metastatic Solid Tumors With EGFR and / or HER2 Aberrations	I

PIK3CB F155C

NCT ID	Title	Phase
NCT04591431	The Rome Trial From Histology to Target: the Road to Personalize Target Therapy and Immunotherapy	II

Clinical Trials Summary (continued)

PIK3CB F155C (continued)

NCT ID	Title	Phase
NCT04192981	A Phase I Study With Expansion Cohort of Concurrent GDC-0084 With Radiation Therapy for Patients With Solid Tumor Brain Metastases or Leptomeningeal Metastases Harboring PI3K Pathway Mutations	I
NCT03213678	NCI-COG Pediatric MATCH (Molecular Analysis for Therapy Choice)- Phase II Subprotocol of LY3023414 in Patients With Solid Tumors	II
NCT04551521	Continuous ReAssessment With Flexible ExTension in Rare Malignancies - CRAFT: The NCT-PMO-1602 Phase II Trial	II
NCT03673787	Ice-CAP: A Phase I Trial of Ipatasertib in Combination With Atezolizumab in Patients With Advanced Solid Tumours With PI3K Pathway Hyperactivation.	I/II
NCT04958226	An Open-label, Fixed-sequence Study to Assess the Effect of Repeated Doses of Capivasertib on the Pharmacokinetics of Oral Midazolam (a CYP450 3A Probe) in Patients With Advanced Solid Tumours Harboring Alterations in the PI3K/AKT/PTEN Pathway	I

Assay Information

Methodology, Scope, and Application: The OCAv3 next-generation sequencing (NGS) assay identifies more than 3000 annotated mutations of interest (MOIs) broadly categorized into 5 mutation types: single nucleotide variants (SNVs), small insertions/deletions (Indels), large (>3 bases) insertions/deletions (Large Indels), copy number variants (CNVs), and gene fusions. The assay utilizes the Thermo Fisher Scientific Oncomine® Comprehensive Assay v3.0 (OCAv3), a next-generation sequencing (NGS) assay that utilizes a multiplex polymerase chain reaction (PCR) with DNA and RNA extracted from formalin-fixed tissue for sequencing on the Ion S5 XL platform and analyzed by the current version of Torrent Suite Software and Ion Reporter. The OCAv3 NGS Assay currently can reliably identify the presence or absence of >3000 known mutations and polymorphisms in the 161 unique genes compared to the Human Reference Genome hg19, including the genes listed below. The OCAv3 assay is a laboratory developed test designed to find gene mutations within tumors (somatic mutations).

Analytical Sensitivity and Specificity: The OCAv3 NGS Assay has been determined to be suitably analytically sensitive and specific for the various types of abnormalities within its reportable range, demonstrating 97.18% sensitivity compared with orthogonal assay results and 100% specificity for the 5 mutation types reported. 99.99% or greater reproducibility has been demonstrated. All quality measures for this assay were within defined assay parameters.

Hereditary and Germline Mutations: This assay examines tumor tissue only and does not examine normal (non-tumor) tissue. Mutations detected by the assay may be present only in the tumor, or in every cell of the body (including non-tumor cells). This test also cannot tell whether a potential germline mutation causes or will cause a hereditary cancer syndrome. If the patient's history includes any one or combination of features suggestive of a possible hereditary cancer predisposition (such as, cancer arising at a young age, especially a cancer of a type unusual in younger patients; a personal history of multiple different types of cancers; or a significant cancer history among blood relatives, especially but not exclusively of the same type as the patient's) it is recommended that the physician arrange for the patient to meet with a genetic counselor and, if warranted, undergo the appropriate genetic test on normal (i.e. non-tumor) tissue (blood or cells brushed from the oral surface of the cheek) to check for a germline abnormality, regardless of the results of this research study.

Report Content: Thermo Fisher Scientific's Ion Torrent Oncomine Reporter software was used in the generation of this report. Software was developed and designed internally by Thermo Fisher Scientific. The analysis was based on the current version of Oncomine Reporter. The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. FDA information was sourced from www.fda.gov and is current as of this version. NCCN information was sourced from www.nccn.org and reflects its current version. Clinical Trials information reflects the current version. For the most up-to-date information regarding a particular trials, search www.clinicaltrials.gov by NCT ID or search local clinical trials authority website by local identifier listed in 'Other Identifiers.' Variants are reported according to HGVS nomenclature and classified following AMP/ASCO/CAP guidelines (Li et al.

2017). Based on the data sources selected, variants, therapies, and trials identified in this report are listed in order of potential clinical significance but not for predicted efficacy of the therapies.

Genes assayed for hotspot mutations:

AKT1, AKT2, AKT3, ALK, AR, ARAF, AXL, BRAF, BTK, CBL, CCND1, CDK4, CDK6, CHEK2, CSF1R, CTNNB1, DDR2, EGFR, ERBB2, ERBB3, ERBB4, ERCC2, ESR1, EZH2, FGFR1, FGFR2, FGFR3, FGFR4, FLT3, FOXL2, GATA2, GNA11, GNAQ, GNAS, H3F3A, HIST1H3B, HNF1A, HRAS, IDH1, IDH2, JAK1, JAK2, JAK3, KDR, KIT, KNSTRN, KRAS, MAGOH, MAP2K1, MAP2K2, MAP2K4, MAPK1, MAX, MDM4, MED12, MET, MTOR, MYC, MYCN, MYD88, NFE2L2, NRAS, NTRK1, NTRK2, NTRK3, PDGFRA, PDGFRB, PIK3CA, PIK3CB, PPP2R1A, PTPN11, RAC1, RAF1, RET, RHEB, RHOA, ROS1, SF3B1, SMAD4, SMO, SPOP, SRC, STAT3, TERT, TOP1, U2AF1, XPO1

Genes assayed with full exon coverage:

ARID1A, ATM, ATR, ATRX, BAP1, BRCA1, BRCA2, CDK12, CDKN1B, CDKN2A, CDKN2B, CHEK1, CREBBP, FANCA, FANCD2, FANCI, FBXW7, MLH1, MRE11, MSH2, MSH6, NBN, NF1, NF2, NOTCH1, NOTCH2, NOTCH3, PALB2, PIK3R1, PMS2, POLE, PTCH1, PTEN, RAD50, RAD51, RAD51B, RAD51C, RAD51D, RB1, RNF43, SETD2, SLX4, SMARCA4, SMARCB1, STK11, TP53, TSC1, TSC2

Genes assayed for copy number variations:

AKT1, AKT2, AKT3, ALK, AR, AXL, BRAF, CCND1, CCND2, CCND3, CCNE1, CDK2, CDK4, CDK6, EGFR, ERBB2, ESR1, FGF19, FGF3, FGFR1, FGFR2, FGFR3, FGFR4, FLT3, IGF1R, KIT, KRAS, MDM2, MDM4, MET, MYC, MYCL, MYCN, NTRK1, NTRK2, NTRK3, PDGFRA, PDGFRB, PIK3CA, PIK3CB, PPARG, RICTOR, TERT

Genes assayed for detection of fusions:

AKT2, ALK, AR, AXL, BRAF, BRCA1, BRCA2, CDKN2A, EGFR, ERBB2, ERBB4, ERG, ESR1, ETV1, ETV4, ETV5, FGFR1, FGFR2, FGFR3, FGR, FLT3, JAK2, KRAS, MDM4, MET, MYB, MYBL1, NF1, NOTCH1, NOTCH4, NRG1, NTRK1, NTRK2, NTRK3, NUTM1, PDGFRA, PDGFRB, PIK3CA, PPARG, PRKACA, PRKACB, PTEN, RAD51B, RAF1, RB1, RELA, RET, ROS1, RSPO2, RSPO3, TERT

DISCLAIMER

This assay is considered a Laboratory Developed Test (LDT). Its performance characteristics have been determined through extensive testing although it has not been cleared by the US Food and Drug Administration and such approval is not required for clinical implementation. Furthermore, any comments included in the report are strictly interpretive and the opinion of the reviewer. This laboratory is certified under the Clinical Laboratory Improvements Amendments (CLIA) for the performance of high-complexity testing for clinical purposes. The correlative data presented is a result of the curation of published data sources, but may not be exhaustive. The content of this report has not been evaluated or approved by the FDA or other regulatory agencies.

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